Computational Identification of High-Altitude Molecules Using Molecular Networking and Liquid Chromatography High-Resolution Mass Spectrometry (LC-HRMS): Developing Software by Chemists, for Chemists

Independent Study Thesis

Presented in Partial Fulfillment of the Requirements for
the Degree Bachelor of Arts in the
Departments of Chemistry & Computer Science at The College of
Wooster

by
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The College of Wooster
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Abstract

Include a short summary of your thesis, including any pertinent results. This section is *not* optional, and the reader should be able to learn the meat of your thesis by reading this (short) section.

This work is dedicated to the future generations of Wooster students.

Acknowledgments

No part of my journey at The College of Wooster has been accomplished alone. Behind every class, project, difficult conversation, and moment of growth are people who have supported me, challenged me, and reminded me why I started. This Independent Study is no exception. I am grateful to everyone who helped me become not only a better student and researcher, but also a better person.

First, I want to thank my Senior Independent Study advisors, Dr. Faust and Dr. Musgrave, for guiding me through a project that brings chemistry and computer science together. Dr. Faust, you have been one of my biggest champions. Thank you for believing in me, advocating for me, and helping me see what I am capable of even when I have struggled to see it myself. Dr. Musgrave, thank you for your guidance, patience, and willingness to work through the challenges of turning an interdisciplinary idea into a meaningful computer science project. I am also grateful to Dr. Paul Edmiston for bringing the original molecular-networking problem to my attention and giving me a reason to explore how the two disciplines I love could work together.

To the Computer Science Department, thank you for teaching me how to think critically, build things, and persist when the first solution fails. Dr. Heather Guarnera, thank you for always helping me find a way to make the impossible seem possible. Dr. Drew Guarnera, as my first computer science professor, you helped build a foundation that continues to shape my work. Dr. Palmer, thank you for contributing to my growth as a programmer and computer scientist. I am grateful to the rest of the department as well, including the faculty and staff whose time, encouragement, and teaching helped me find my place in the field.

To the Chemistry Department, thank you for making science something I wanted not only to study, but to spend my life exploring. Dr. Martin, thank you for your guidance as my academic advisor and for supporting my path across two majors. Dr. Snider, thank you for the difficult but necessary conversation that helped me grow. Dr. Bonvallet, thank you for making lectures something to look forward to. Dr. Gallo, thank you for helping me fall in love with chemistry. Dr. Baker, thank you for being one of my favorite professors and one of my favorite people to stop and say hello to. Dr. Feierabend, thank you for pushing me to be great, especially in the moments when I did not fully believe I could be. I am grateful to every other member of the Chemistry Department who shared their knowledge, offered encouragement, or made the department feel like a community.

To my best friends, Kate Carson, Eliza Davis, and Brooke Bradley, thank you for being part of my life through the busy days, the stressful weeks, and the moments worth celebrating. Your friendship has meant more to me than I can put into a single paragraph. Thank you for giving me people I could laugh with, rely on, and be myself around.

To my brothers of Beta Kappa Phi, thank you for giving me a brotherhood and a home within Wooster. I especially want to recognize Shea McLaren, Cooper Norwell, Gabriel Jones, Nathan Krieg, Parth, Isaac Prince, Jackson Ferrell and Matthew Dalton, my littles, Hayden Jacoby and Peter Dentzer, my grand big and big. Each of you has helped shape what brotherhood, responsibility, and friendship mean to me. I am proud of the memories we have made and the people we have become together. The first and finest will always hold a special place in my heart.

To the Residence Life Office, thank you for the opportunity to grow as a leader and learn what it means to show up for a community. I also want to thank Maddie Cuckow, my former AD and now Greek Life Director, for being part of that journey and for the support and guidance you have given me along the way.

I want to give a special thank-you to Lauren Richmond, who has been an important part of my life and personal growth throughout the years. Although life has taken us in different directions, I will always be grateful for the love, encouragement, and support she has given me. The memories we created, the challenges we faced, and the experiences we shared have shaped me in ways I will always appreciate. Her impact on my journey extends far beyond our time together, and I will always value the role she played in helping me become the person I am today.

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To Fr. David, thank you for walking alongside me during my journey toward baptism. Your guidance helped me through a deeply personal part of my life, and I am grateful for the time, care, and faith you shared with me.

Most importantly, I thank my family and siblings, especially my seven sisters and six brothers. You are part of the reason I care so deeply about using what I learn to make a difference. Thank you for the love, sacrifices, patience, and encouragement that have carried me through every stage of my education. Whatever I accomplish, I will always carry my family with me.

Finally, to everyone whose name may not appear here but who has offered a kind word, a difficult lesson, a new opportunity, or a reason to keep going, thank you. This work may have my name on its cover, but the person who completed it was shaped by all of you.

Vita

Raelyn M. Brooks is a candidate for the Bachelor of Arts degree in Chemistry and Computer Science at The College of Wooster, with an anticipated graduation date of May 2027. His academic interests lie at the intersection of computational chemistry, software engineering, and scientific research. He is particularly interested in developing computational tools that allow scientists to investigate complex chemical problems more efficiently and reliably.

During his undergraduate studies, Brooks pursued interdisciplinary research combining chemistry and computer science. His Senior Independent Study, advised by Dr. Jennifer Faust and Dr. John Musgrave, focuses on developing molecular-networking software to support the investigation of unknown molecular features in remote environmental samples using liquid chromatography–high- resolution mass spectrometry (LC-HRMS). His previous research includes developing a Python-based cheminformatics pipeline using RDKit to explore multistep organic synthesis pathways across large chemical reaction datasets.

Brooks has gained professional experience in software engineering and cloud computing. As a Software Developer Intern at the Federal Reserve Bank of St. Louis, he contributed to production software development, automated infrastructure, and security remediation. He has also worked as an Associate Cloud Engineer at GoHub LLC, developing cloud infrastructure and software for a transportation platform. Additionally, he has served as a coding instructor at iD Tech, teaching programming, software development, and problem-solving to hundreds of students.

Outside the classroom, Brooks has been actively involved in campus leadership and community engagement. He has served as President of Beta Kappa Phi, participated in Black & Code, and contributed to student life through Residence Life and other campus organizations. These experiences have strengthened his commitment to mentorship, leadership, and creating opportunities for others.

Following graduation, Brooks intends to pursue a career in software engineering, particularly in developing computational tools for chemistry and scientific research. He also hopes to pursue graduate study in chemistry, computer science, or an interdisciplinary field connecting the two disciplines.

Fields of Study Major field: Chemistry & Computer Science

Specialization: Computational Chemistry, Molecular Networking, and Liquid Chromatography–High-Resolution Mass Spectrometry (LC-HRMS)

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Preface

I have always considered myself a problem solver. Whether I am writing code, working through a chemical reaction, or trying to understand why something does not work as expected, I am drawn to problems that require patience, creativity, and a willingness to approach things differently. This is ultimately what attracted me to both chemistry and computer science. Chemistry has taught me to think beyond what I can immediately observe and to consider the countless interactions that explain the world around us. Computer science has taught me that failure is often part of progress, and that solving a problem sometimes requires starting over, questioning an assumption, or finding an entirely different approach. Together, these disciplines have shaped how I think, how I learn, and the kind of work I hope to pursue.

The inspiration for this Independent Study came from a conversation with Dr. Paul Edmiston, who introduced me to a challenge involving molecular-networking software. His students were exploring how molecular networking could help identify chemically modified substances relevant to forensic toxicology and support investigations involving drug-facilitated crimes. However, understanding the computational processes behind the software and evaluating the reliability of its results presented a challenge. Dr. Edmiston needed someone willing to look beneath the surface of the software and understand how it worked. As a brother to seven sisters, the opportunity to contribute to research with potential implications for personal safety and justice resonated with me deeply. Beyond its scientific importance, I recognized a chance to apply my knowledge of chemistry and computer science to a problem that could make a meaningful difference in people's lives.

What made this opportunity especially exciting was that it brought together the two fields I had spent my undergraduate career studying. I have never viewed chemistry and computer science as competing interests. Instead, I see them as disciplines that strengthen one another. Chemistry presents complicated questions about the world, while computer science provides tools for investigating those questions at a scale and level of detail that would otherwise be difficult to achieve. Although the research eventually developed into a project centered on molecular networking and the identification of unknown features in remote environmental samples, that initial conversation remained central to my motivation: I wanted to understand how the software worked and how it could be made more useful and reliable for the scientists who depend on it.

My experiences outside the classroom have further shaped that goal. During my internship as a software developer at the Federal Reserve Bank of St. Louis, I learned the importance of writing dependable software, testing my work, and understanding the consequences of decisions made within complex systems. Working with cloud infrastructure and software architecture has shown me how careful design affects reliability and long-term usefulness. As a coding instructor, I have taught hundreds of students how to approach difficult programming problems, often reminding them of the same lesson I continue to learn myself: not understanding something immediately does not mean it cannot be understood. My previous computational chemistry work, including developing tools to explore multistep organic synthesis using large reaction datasets, also helped me recognize how powerful software can become when it is built with a clear understanding of the science it supports.

After graduation, I hope to pursue a career developing software for chemists and other scientific researchers. I want to build tools that do more than produce results. I want researchers to understand those results, explore the evidence behind them, and have confidence in the computational processes they use. This Independent Study represents my first substantial attempt to bring that ambition into practice. It has also challenged me to recognize that being both a chemist and a computer scientist does not mean having every answer. Instead, it means being willing to learn from both disciplines, recognize the limitations of each, and do the work necessary to connect them.

I would not have reached this point without the faculty members who have challenged, supported, and believed in me throughout my time at The College of Wooster.

Within the Computer Science Department, I want to thank Dr. Heather Guarnera for consistently helping me find ways to make what once seemed impossible feel achievable. Your encouragement has pushed me to think bigger about what I am capable of accomplishing. I thank Dr. Drew Guarnera, my first computer science professor, for helping establish the foundation upon which so much of my work now rests. I am also grateful to Dr. Palmer for contributing to my growth as a computer scientist and to Dr. Musgrave, my Senior Independent Study advisor, for guiding me through the computational challenges of this interdisciplinary project.

Within the Chemistry Department, I owe a special thank-you to Dr. Faust, my Senior Independent Study advisor and one of my greatest champions. Your belief in my abilities, your encouragement, and your willingness to help me grow have meant more than I can adequately express. I thank Dr. Martin, my academic advisor, for helping guide my academic journey and supporting my interests across two demanding disciplines. Dr. Snider, thank you for the difficult but necessary conversation that helped me recognize where I needed to grow. Sometimes the advice we need most is not the advice we expect to hear. I thank Dr. Bonvallet for being a lecturer whose teaching has consistently made learning chemistry engaging. I thank Dr. Gallo for

helping me fall in love with chemistry and recognize just how much there is to discover within it. Dr. Baker, thank you for being one of my favorite professors and for the simple kindness of always being someone I look forward to greeting. Finally, Dr. Feierabend, thank you for continuing to challenge me to pursue excellence, particularly during moments when I have struggled to recognize my own potential.

I also want to thank Dr. Edmiston for presenting the problem that ultimately helped me discover the direction I want to take my career. Sometimes the most important opportunities begin with a conversation and a question that nobody has fully answered.

This Independent Study is more than a requirement for graduation. It represents the intersection of my interests, my experiences, and the people who have helped shape my education. More importantly, it is a beginning. I entered college wanting to understand how things work. I hope to leave having taken my first meaningful steps toward building things that help others understand the world.

CHAPTER 1

Introduction

1.1 Research Problem and Motivation

Anthropogenic chemicals enter the environment through industrial activities, agricultural practices, and the manufacture and use of consumer products. These inputs include pharmaceuticals, pesticides, industrial chemicals, and plastic-associated compounds, among other substances. Some chemicals are sufficiently persistent or mobile to move beyond their immediate sources, complicating efforts to determine their distribution and environmental fate [18, 3, 29]. Their occurrence is not confined to urban or industrial settings. Investigations of mountain regions and remote waters demonstrate that chemical pollution can be detected far from major emission centers, although the relative contributions of long-range transport and local activities differ among sites [7, 31, 17].

The analytical problem extends beyond determining whether a sample contains selected contaminants. Environmental samples may produce many signals in liquid chromatography–high-resolution mass spectrometry (LC-HRMS), but detecting a signal does not establish the chemical identity of the substance responsible for it. Traditional targeted measurements are useful when the compounds of interest are known in advance; they are less suited to surveying chemical features that were not anticipated when a method was developed. Suspect screening and non-targeted analysis (NTA) broaden the range of features that can be investigated, yet incomplete reference libraries and uneven spectral coverage continue to constrain identification [18, 23, 14].

This limitation is especially important for remote environmental samples. A database match may support a candidate assignment for one feature, while nearby features in the same dataset remain unannotated despite containing potentially useful tandem mass spectrometry (MS/MS) information. Molecular networking offers a way to examine these data by representing features as nodes and retaining relationships between features with

sufficiently similar MS/MS spectra. The resulting network can help researchers explore candidate chemical relationships and prioritize features for further study, but a network connection is not itself a confirmed molecular identification [26, 2, 24].

The present Senior Independent Study addresses a second, connected problem: the need for a dependable computational workflow through which chemists can construct, inspect, preserve, and interpret molecular networks. For the College of Wooster's environmental HRMS research, useful software must connect the displayed network to the underlying spectral evidence and feature metadata while making its computational decisions reproducible. This project therefore combines environmental chemistry with software engineering. Its central contribution is the development and evaluation of a molecular-networking workflow and interface, using environmental HRMS data as both the motivating application and the basis for chemical case studies.

1.2 Environmental and Chemical Background

1.2.1 Long-Range Transport of Organic Contaminants

Once released, an organic chemical's distribution depends on its physicochemical properties, its susceptibility to degradation, and the environmental processes acting on it. Atmospheric transport is one route by which certain pollutants can reach distant ecosystems. Volatilization, emissions associated with combustion or other activities, and the movement of particle-associated compounds can introduce chemicals into the atmosphere. During transport, reactions such as oxidation and photolysis may alter the original substances, producing transformation products with properties that differ from those of their precursors [7, 32]. The presence of a chemical in a remote sample, however, does not by itself establish its source or the pathway by which it arrived.

For some semivolatile organic compounds (SVOCs), repeated volatilization, atmospheric movement, and redeposition can contribute to long-distance redistribution. This sequence is often described as the *grasshopper effect*. Global fractionation and related temperature-dependent partitioning processes can favor the transfer of certain chemicals toward cooler regions, including mountain environments [28, 7]. These mechanisms are particularly relevant to persistent compounds whose transport and deposition can occur over extended timescales. They should not be generalized to every pharmaceutical, pesticide, or other chemical detected by HRMS, because environmental behavior varies substantially among substances.

Mountain topography and climate influence where transported contaminants are deposited. Orographic precipitation can increase wet deposition as air masses rise and cool, while snowfall can scavenge some gas-phase and particle-associated chemicals. Temperature gradients also affect air–surface exchange and the

partitioning of SVOCs among air, snow, vegetation, and other surfaces [4, 13, 8]. These processes operate together, rather than as independent explanations for every observation at a mountain site.

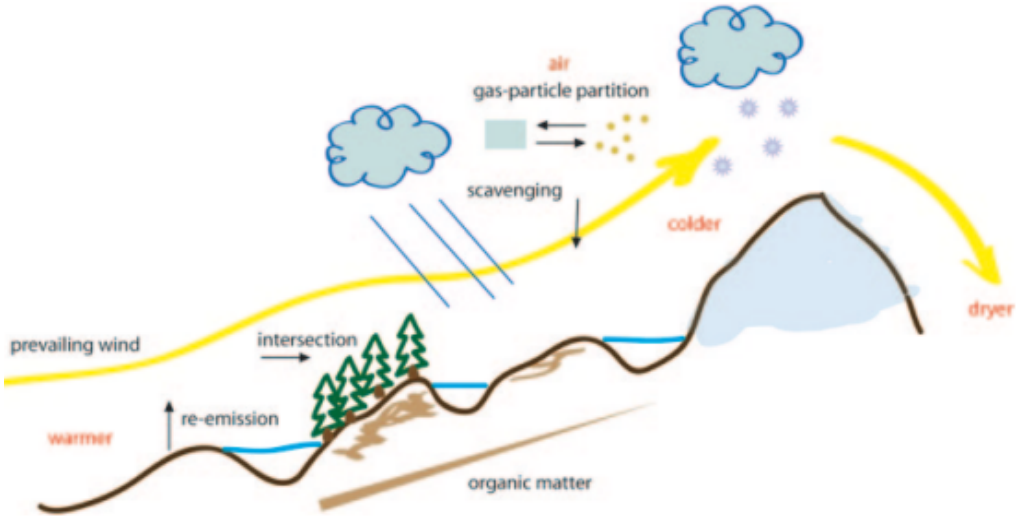


Figure 1.1: Conceptual pathways for atmospheric transport and deposition of contaminants in mountainous terrain, including precipitation scavenging, air–surface exchange, and elevation-dependent environmental conditions. Reproduced or adapted from the source discussed by Catalan [4]; confirm the original figure attribution.

1.2.2 High-Altitude and Remote Surface-Water Environments

High-altitude regions offer valuable settings for examining contaminant transport because their climatic conditions can favor deposition, seasonal storage, and subsequent redistribution of chemicals. Mountain snow and ice may temporarily retain atmospheric contaminants, and melting can transfer some of these substances into downstream waters. The timing and magnitude of release depend on snowpack processes and the properties of the chemicals involved [13, 19]. Such pathways matter for interpreting surface-water samples, but the detection of a feature alone does not establish its ecological effect or the extent of exposure to organisms.

Research on mountainous environments has documented persistent organic contaminants and other human-associated chemicals in regions that are geographically distant from major industrial centers [7, 31, 27]. Studies of the Qinghai–Tibetan Plateau also illustrate that spatial patterns of contamination can be influenced by human activity within a highland basin, not solely by long-range atmospheric transport [17]. Consequently, these environments should be treated as chemically informative rather than assumed to be free of local sources. Tourism, settlement, transportation, and regional activities may contribute alongside atmospheric inputs, depending on the location and compound.

Cold trapping is one explanation for the accumulation of particular semivolatile contaminants in mountainous environments. As air temperatures decrease with elevation, some compounds partition more readily into cooler environmental compartments. Seasonal changes can then modify the balance between deposition and re-emission. These exchanges provide a conceptual explanation for how contaminants can be redistributed; they are not direct measurements of the samples analyzed in this thesis [7, 13].

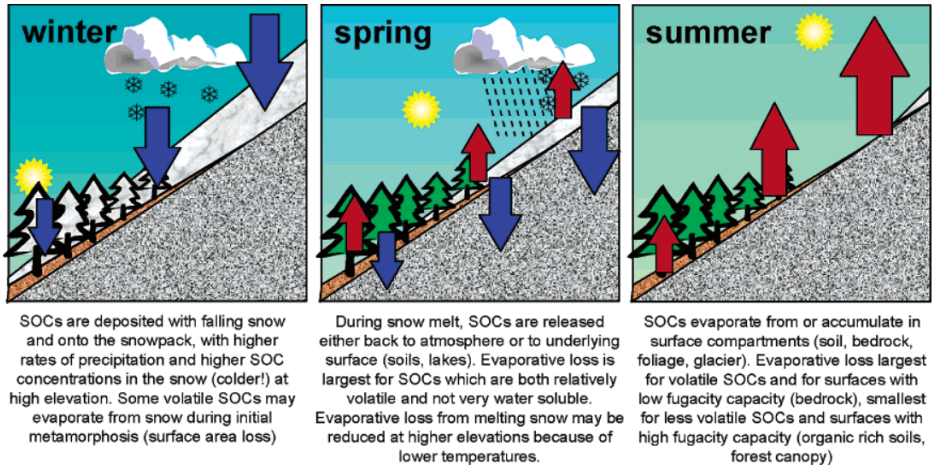


Figure 1.2: Illustration of seasonal air–surface exchange processes affecting semivolatile organic contaminants along a mountain slope. From the discussion of organic contaminants in mountains by Daly and Wania [7]; verify reproduction or adaptation permissions.

The published mountain literature provides context for the environmental samples considered in this project. Daly and Wania reviewed studies of persistent organic contaminants at mountain locations and the factors affecting their transport and retention [7]. The compilation previously reproduced in the Junior I.S. as a figure is useful as evidence of the range of mountain sites and chemical classes investigated; it should not be described as a plot of contaminant concentrations because it summarizes studies and sampling information rather than presenting a common quantitative scale.

The present project is motivated by environmental HRMS datasets associated with remote and high-altitude locations, including the Himalayas, Mount Kilimanjaro, and the Qinghai–Tibetan Plateau. These locations provide a chemical context in which both previously recognized contaminants and unannotated features are of interest. The exact samples, matrices, collection locations, and data subsets included in the software evaluation will be specified in Chapter 2. No claim is made here that every location mentioned in the literature is represented in the dataset used for the final case studies.

chemical	location (elevation[a])	date	ref
	OPPs and Other Currently Used Pesticides		
chlorpyrifos, diazinon, parathion	3 elevations (114/200, 533, 1920 m) in and below Sequoia National Park, Sierra Nevada, CA	15 days (Nov. 1990–Feb. 1991)	*26*
chlorpyrifos, methidathion		8 days (May–Oct. 1994)	*27*
chlorpyrifos, diazinon, malathion, chlorothalonil, trifluralin, endosulfan		5 days (May–Sept. 1996)	*28*
	SOCs		
PCBs in 2 HiVol samples	Lake Tahoe, Sierra Nevada, CA, USA (ca. 1900 m)	July 1995	*29*
OCPs in 7 HiVol samples	Bow Lake, Rocky Mtns., Canada (1940 m)	summer 1998	*61*
PAHs in 36 HiVol samples	Estany Redó, Pyrenees (2240 m)	1996–1997	*34*
	Gossenköllesee, Austrian Alps (2413 m)	1996–1997	
	Øvre Neådalsvatn, Norway (728 m)	1998	
PCBs and OCPs in 23 HiVol samples	Bow Lake (1974 m), Donald Station (770 m), Lower Kananaskis Lake (1667 m), Sundre, Rocky Mountains, Canada (1115 m)	June 1999–Aug. 2000	*33*
PCBs and OCPs in 20 HiVol samples	Pico de Teide, Canary Islands (47 and 2367 m)	June 1999–July 2000	*25*
α-HCH, HCB, DDT	Yahiko Mtn. (634 m), Japan	May–Nov. 2001	*30*
OCPs in 8 passive samples	Rocky Mountains, Canada (800–2300 m)	2000–2001 (annual average)	*31, 32*
PCBs and OCPs in 30 HiVol samples	Estany Redon, Central Pyrenees (2240 m)	Nov. 2000–Jan. 2003	*106*
	Skalnate Pleso, High Tatras (1778 m)		
OCPs in 6 HiVol samples	Dingri (4340 m), Rongbu Valley (5050 m), Tibetan Plateau	May–June 2002	*16*

[a] Elevation given in meters above sea level.

Figure 1.3: Summary of selected mountain-air contaminant investigations, including reported chemical classes, study locations, and sampling periods, as compiled by Daly and Wania [7]. This is a study-summary table, not a comparison of measured concentrations. Verify the original figure attribution.

1.2.3 Previous Wooster Research and the Identification Gap

Previous work associated with the College of Wooster provides a direct foundation for this study. Fan's undergraduate research investigated suspect screening and semi-quantification of organic contaminants in snow and water from the Qinghai–Tibet Plateau and other high-mountain regions [10]. Related research involving Wooster investigators has applied LC-HRMS suspect screening to pesticides in rain and snow, as well as NTA and suspect screening to organic contaminants in temperate snowfall [12, 20]. These studies are not interchangeable: the temperate precipitation investigations demonstrate analytical approaches, while Fan's work is more directly connected to the remote high-altitude samples motivating this project.

The identification gap arises because feature detection can yield more information than reference-based annotation can explain. Accurate mass, retention time, isotope information, and MS/MS spectra can be available for a detected feature even when its chemical structure remains uncertain. An unannotated feature may represent a chemical absent from the library, a feature with insufficient-quality evidence, or another analytical signal requiring additional interpretation. It should not automatically be described as a new contaminant or a transformation product [18, 23, 24].

This distinction defines the need for both chemistry and computer science in the current work. From the chemistry perspective, researchers need to organize evidence and prioritize features that merit further investigation. From the computational perspective, they need to calculate spectral relationships consistently,

retain the information behind those relationships, and navigate the resulting network without losing the connection to the original data. The purpose of the new software is to support this investigative process, not to replace confirmatory chemical analysis.

1.3 LC-HRMS, Suspect Screening, and Non-Targeted Analysis

1.3.1 LC-HRMS and Tandem Mass Spectrometry

Liquid chromatography separates components of a sample before mass-spectrometric detection, helping resolve some of the complexity of environmental mixtures. High-resolution mass spectrometry measures ions with sufficient mass accuracy to support candidate elemental compositions and other forms of chemical investigation. Depending on the acquisition and processing methods, an LC-HRMS feature can be characterized by a mass-to-charge ratio (m/z), retention time, signal intensity, and additional evidence such as an isotope pattern. These attributes are useful for organizing a dataset, but they do not necessarily determine a unique chemical structure [18, 16].

Tandem mass spectrometry provides another source of information by selecting precursor ions and recording the product ions generated during fragmentation. MS/MS patterns can assist in distinguishing candidates, comparing features, and searching spectral libraries. Their interpretation depends on acquisition conditions, spectral quality, and the reference data available for comparison [16, 14]. The specific LC-HRMS instrumentation, ionization modes, acquisition strategy, and upstream processing steps associated with the datasets used in this thesis are documented in Chapter 2 rather than assumed from the general analytical literature.

1.3.2 Molecular Features Versus Identified Compounds

A *molecular feature* is an analytical observation rather than a confirmed chemical identity. Multiple observed signals can arise from a single substance through different adducts, isotopes, or fragments, while different substances can also produce overlapping or difficult-to-distinguish signals. For this reason, a feature count is not equivalent to a count of unique contaminants. Likewise, a spectral library match or a relationship to an annotated neighbor does not automatically justify calling an unknown feature an identified compound [18, 24].

Confidence in a chemical assignment should be communicated in relation to the supporting evidence. Schymanski and colleagues proposed a framework for reporting confidence levels in HRMS-based small-molecule identification, ranging from confirmed structures to less specific candidate or formula-level information [24].

Within this thesis, *feature* is used when identity has not been established, while *candidate annotation* or *tentative identification* describes an assignment supported by incomplete evidence. A confirmed identification requires the appropriate confirmation criteria, such as comparison with an authentic reference standard when relevant. The confidence associated with individual case-study features will be stated rather than inferred from network appearance.

1.3.3 Suspect Screening and Non-Targeted Analysis

Targeted analysis begins with a defined set of analytes and methods selected to measure them. It is valuable for questions concerning known chemicals, but its scope is constrained by the compounds included in the method. Suspect screening broadens that scope by examining acquired data for chemicals of prior interest, often using information from candidate lists and available analytical evidence. NTA takes a wider approach by retaining and investigating features without requiring that every candidate be specified before data acquisition [18, 23].

These strategies are complementary rather than mutually exclusive. Suspect screening can prioritize features associated with known environmental concerns, while NTA can reveal additional signals that do not match the current suspect list. Published studies of pesticides in rain and snow and of broader organic chemical mixtures in snowfall demonstrate the value of these approaches in environmental LC-HRMS workflows [12, 20]. At the same time, an expanded set of detected features creates a larger interpretation problem: researchers must decide which signals are chemically meaningful and what evidence supports each possible assignment.

The term *exposome* provides a broader context for this challenge by referring to the totality of exposures across time, including chemical exposures. NTA and suspect screening can contribute information about portions of this chemical landscape, but no single analytical dataset measures the entire exposome [18]. For the present project, the more immediate objective is to investigate environmental feature relationships in the available data, not to quantify total exposure or infer health effects.

1.3.4 Limitations of Current Identification Approaches

Even when high-resolution data are available, many features remain without confident annotations. Reference databases may not contain a relevant chemical or its spectrum, experimental conditions may differ from those used to generate a library spectrum, and the available MS/MS data may be incomplete or ambiguous. Reviews of environmental NTA and computational structural annotation describe these constraints and emphasize the importance of quality control and careful reporting [23, 14, 11].

The consequence is a gap between *detection* and *interpretation*. Increasing the number of observed features

does not necessarily increase the number of confidently identified substances. Additional computational methods can help organize the evidence, identify promising comparisons, and prioritize follow-up work, but they must preserve uncertainty instead of presenting every computational association as a chemical conclusion. Molecular networking is one such method, and its theoretical basis is developed in the following section.

1.4 Molecular Networking and Theoretical Foundations

1.4.1 MS/MS Fragmentation and Spectral Similarity

A tandem mass spectrum records product ions and their intensities after a precursor ion undergoes fragmentation. Structurally related molecules can share fragments or fragmentation pathways, which creates a basis for comparing their spectra. Nevertheless, the relationship between structure and spectral similarity is not one-to-one. Different compounds may share characteristic fragments, and the same compound may yield different relative intensities under different acquisition conditions. Spectral comparisons are therefore useful evidence for exploratory analysis, not direct proof of identity [26, 2].

In a molecular network, this comparison becomes a relationship between features. An annotated feature may provide context for neighboring features, including those that have no direct library match. Such proximity can help prioritize a chemical family or candidate structure for further investigation, but any proposed relationship must be evaluated against the available fragmentation data, precursor information, and independent chemical evidence [22, 9, 24].

1.4.2 Cosine Similarity and Molecular-Network Construction

Cosine similarity is a common method for measuring agreement between represented mass spectra. After an appropriate peak-matching or alignment procedure, spectra can be expressed as nonnegative intensity vectors $\mathbf{x}$ and $\mathbf{y}$. Their cosine similarity is

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \, \|\mathbf{y}\|_2}. \tag{1.1}$$

For nonnegative intensity vectors, values closer to one indicate greater agreement under the selected representation and matching procedure. The score is meaningful only in the context of those choices. Peak matching, mass tolerances, intensity treatment, minimum matching requirements, and any modified-cosine or neutral-loss logic can change the comparisons and the resulting network [2, 6].

The score is not itself an edge. A molecular-networking workflow applies additional rules to determine which pairwise similarities are retained as connections, such as a minimum score or other filters. Different rules can produce different network densities and connected components from the same spectra. Consequently,

a reproducible workflow must document its comparison method and parameter values. Chapter 2 describes the actual algorithm and edge-selection rules implemented in this project; Equation 1.1 establishes the general mathematical foundation rather than claiming that a particular implementation has already been validated.

1.4.3 Graph Theory: Nodes, Edges, and Connected Components

A molecular network can be represented as a graph $G = (V, E)$, where V is the set of nodes representing analytical features and E is the set of retained spectral relationships. An edge may carry a similarity score or other metadata describing the evidence for the relationship. This representation provides a computational structure for organizing large sets of pairwise comparisons and exploring groups of related observations [1, 26].

A *connected component* is a maximal set of nodes linked through paths in the graph. Components can help organize a network into smaller groups for inspection, while isolated nodes represent features without retained edges under the chosen criteria. These are graph properties, not chemical classifications. In particular, two features in the same component may be connected only indirectly, and membership in a component does not establish a shared molecular formula, common source, or transformation pathway.

1.4.4 Network Visualization and Interpretation

Visualization translates the graph into an interface that chemists can explore. A useful view should allow researchers to identify features and their connections, inspect similarity scores, and retrieve the metadata or spectra associated with a selected node or edge. Color, size, layout, and filtering can make patterns easier to see, but these visual choices must be defined so that they do not imply greater chemical certainty than the underlying data justify. Research on interactive graph exploration supports the importance of connecting overview visualizations with inspectable data and user tasks [21, 15].

The distinction between a graphical display and a scientific workflow is important for this project. A static image of a network may communicate broad structure, but it cannot by itself establish which source records produced an edge or whether an analysis can be repeated with the same parameters. The proposed interface therefore emphasizes the relationship between visual elements, underlying feature data, and the computations used to generate connections. The implemented interactions and their evaluation will be described in Chapters 2 and 3.

1.4.5 Limitations of Spectral Similarity as Evidence of Chemical Identity

Spectral similarity can support hypotheses about structural relationships, but it cannot independently confirm the identity of an unknown compound. Shared fragment ions may arise from different structures, and changes in collision energy, ionization, and spectral quality can alter the apparent similarity between spectra [2, 14]. Likewise, a mass difference between two connected features does not, without further evidence, demonstrate that one is a transformation product of the other.

This limitation shapes both the software design and the chemistry case studies. The network should make comparisons inspectable and preserve annotation confidence, while interpretation should distinguish observations, tentative hypotheses, and confirmed assignments. The intended benefit is more informed prioritization of unknown features, not an automatic conversion of spectral relationships into definitive chemical identifications [24, 5].

1.5 Existing Software and Research Gap

1.5.1 MS-DIAL and GNPS in the Existing Workflow

Mass-spectrometry analysis typically involves multiple computational stages. In the Wooster research context, MS-DIAL is used for upstream data processing and feature-level outputs, while molecular-networking approaches associated with the Global Natural Products Social Molecular Networking platform (GNPS) provide methods for organizing MS/MS relationships [25, 26, 22]. These tools address different analytical tasks and have established value in their respective areas. The objective of this thesis is not to suggest that the existing platforms are universally inadequate, nor to replace every stage of LC-HRMS data processing.

Instead, the project focuses on the transition from processed analytical data to a molecular network that can be explored and reproduced in the local research workflow. The input and export formats, metadata requirements, and comparison methods at this boundary affect whether the results remain interpretable after they leave the original processing software. The specific versions and settings used for the project data will be documented in Chapter 2.

1.5.2 Documented Limitations in the Wooster Research Workflow

The software-development motivation arose from difficulties reported in local molecular-networking work, particularly around saving, retaining, or exporting analyses and results. Such problems matter because a network is useful for research only when its underlying data, parameter choices, and interpretations can be

revisited. These concerns are presented as local workflow requirements that motivate this project, not as evidence that GNPS necessarily behaves the same way for every user or deployment.

A reliable workflow must also support the scientific interpretation of its outputs. A chemist examining a connected pair of features needs access to the corresponding identifiers, spectra, annotations, and similarity evidence. Without those links, an attractive visualization may be difficult to verify or use in subsequent analysis. Reproducibility likewise depends on retaining input provenance, computational parameters, and sufficient documentation to repeat the workflow [30, 21]. The particular requirements adopted for this project will be made explicit and evaluated rather than assumed to have been satisfied.

1.5.3 The Need for Software Designed Around Chemists' Needs

The research gap is therefore both analytical and computational. Incomplete reference coverage leaves many environmental HRMS features unidentified, while a practical molecular-networking workflow must make spectral comparisons, graph relationships, and supporting metadata accessible to researchers. Existing tools demonstrate the value of molecular networking, but the requirements of a particular laboratory workflow still need to be defined, implemented, and tested [26, 22, 5].

This thesis approaches software as a scientific research instrument rather than merely a way to display a network. The proposed contribution is a workflow and interface tailored to Wooster's environmental HRMS context, with attention to computational correctness, reproducibility, data handling, and the interpretability of network relationships. Its value will be assessed through software evaluation and chemical case studies. The chemistry contribution concerns what can reasonably be learned or prioritized from the environmental data; the computer science contribution concerns how reliably and transparently the software supports that investigation.

1.6 Research Question and Specific Aims

The central research question is:

> What computational workflow and interface are needed to make molecular networking reliable, reproducible, and useful for Wooster's environmental HRMS research?

The project addresses this question through four aims:

1. **Define the chemistry and data requirements.** Identify the analytical data, metadata, and evidence needed to construct and interpret molecular networks for the environmental HRMS research context.

2. **Develop an end-to-end molecular-networking workflow.** Implement a workflow that processes supported input data, compares MS/MS spectra, constructs a network, and allows users to inspect network elements and associated chemical information.

3. **Evaluate computational correctness, reproducibility, performance, and workflow usefulness.** Test calculations, input handling, graph construction, repeatability, performance, and relevant user-facing tasks using documented evaluation methods.

4. **Apply the software to environmental HRMS case studies.** Use selected environmental data to examine annotated and unidentified features, describe relationships supported by the evidence, and identify remaining uncertainties.

1.7 Project Scope and Thesis Contributions

1.7.1 Project Scope

This project focuses on computational support for molecular networking using existing environmental HRMS data. It includes defining data requirements, implementing and testing the network workflow, designing an interface for inspecting features and spectral relationships, and applying the software to selected chemical case studies. The specific input formats, data-processing assumptions, software features, and evaluation procedures will be defined in Chapter 2.

The project does not assume that a network alone can confirm the identity of unknown compounds, prove chemical transformation pathways, or replace reference standards and other confirmatory methods. Software development alone also does not establish the environmental source, toxicity, or ecological effect of a detected feature. Claims in the chemical case studies will be limited to what the available analytical data and supporting evidence justify.

1.7.2 Expected Computer Science Contributions

The Computer Science contribution is the design, implementation, and evaluation of a molecular-networking workflow and interface for the stated research context. The contribution will be documented through the software architecture, similarity and graph-construction methods, interface behavior, input validation, testing, and reproducibility practices. Chapter 3 will report what was implemented and how it performed under the evaluation procedures.

1.7.3 Expected Chemistry Contributions

The Chemistry contribution is the application of the workflow to environmental HRMS data to investigate spectral relationships among annotated and unidentified molecular features. The case studies will describe the evidence available for selected features, explain how network relationships may help prioritize further investigation, and state the remaining limitations. Tentative network relationships will not be presented as confirmed chemical identifications.

1.7.4 Organization of the Thesis

Chapter 2 describes the research design, environmental datasets and analytical methods, upstream processing, software requirements and implementation, network construction, and evaluation procedures. Chapter 3 presents the dataset summary, software implementation results, computational validation, performance and workflow evaluation, network characteristics, and chemical case studies. Chapter 4 interprets the computational and chemical findings, discusses design trade-offs and limitations, and considers implications for future environmental HRMS research. Chapter 5 summarizes the contributions, answers the central research question, and identifies future directions.

CHAPTER 2

Methodology, Research Design, and Software Development

2.1 Overall Research Design and Workflow

2.2 Environmental Samples and Chemistry Experimental Methods

2.3 Upstream Data Processing and Chemical Annotation

2.4 Software Requirements and System Architecture

2.5 Spectral Comparison and Similarity Calculation

2.6 Molecular-Network Construction

2.7 User Interface Design and Interaction

2.8 Software Implementation and Reproducibility

2.9 Software Evaluation and Testing Methods

2.10 Chemical Case-Study Design

CHAPTER 3

Results and Data Analysis

3.1 Environmental Dataset and Feature Summary

3.2 Software Implementation Results

3.3 Computational Correctness and Validation

3.4 Software Performance and Workflow Evaluation

3.5 Molecular-Network Structure and Characteristics

3.6 Chemical Case Study 1: Annotated Molecular Features

3.7 Chemical Case Study 2: Unidentified Molecular Features

3.8 Summary of Principal Results

CHAPTER 4

Discussion

4.1 Interpretation and Integration of Results

4.2 Computational Contributions and Design Trade-Offs

4.3 Chemical Significance and Comparison with Literature

4.4 Study and Implementation Limitations

4.5 Implications for Future Environmental HRMS Research

CHAPTER 5

Conclusions and Future Work

5.1 Summary of the Research

5.2 Principal Computer Science Contributions

5.3 Principal Chemistry Contributions

5.4 Answer to the Central Research Question

5.5 Future Work

5.6 Final Remarks

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